

To Go Boldly Where No Board Has Gone Before



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Building projects for flight and space is one of the most popular areas in recent times. From building mini satellites to setting up weather balloons that fly to what is termed as near space, open source hardware and software solutions are driving the poor man's space program.

"Learning to develop engineering skills is like learning to lift weights. You should start with an empty exercise and then keep adding challenges," explains Eben Upton, founder of the Raspberry Pi Foundation, in a recent interview on the eve of Raspberry Pi 2 launch. Open source hardware tools are built to do just that.

Arduino, one of the earliest and most popular lines of open source hardware tools, has had a growing toolset since 2005. Having withstood the test of time, it now enjoys a passionate following of engineers, tinkers and hackers who use it to prototype products and test project designs. In 2015, we now have a vast collection of open hardware developer platforms to choose from.

Over the past decade, we have seen open hardware being used for a variety of projects, including fun projects like building a radio-controlled (RC) car, utilitarian projects like implementing a home media centre or server, all the way to setting up serious industrial automation systems that can control a building. But is that all?

Up above the world so high

Aerospace. That is where the excitement lies these days. During the interview, Upton talked about his favourite kind of project. "I am a big fan of space-style projects, and Dave Akerman is the one who has been doing some fascinating work in sending Raspberry Pi up in weather balloons." The latest iteration of the device features radio telemetry, slow scan digital video (SSDV) images,

a 3G link for video and backup telemetry. It sends a live video stream as well.

Aerospace projects are no more something that governments alone can do. For instance, SpaceX and Elon Musk have made the sector hot from both a commercial and passion standpoint, with its ultimate goal stated to be that of enabling people to live on other planets.

Would you like to...

Let us take a look at how open source solutions have helped build cool aerospace projects, and how you can leverage the same for your aerospace project.

... perform space experiments

Setting up experiments that can be run in the International Space Station (ISS) is not far-fetched anymore. Explore some of these cool open hardware solutions that will let you do just the thing.

ArduSat

This is probably one of the most famous projects to have taken the humble Arduino and ridden it to space. It is built to the specifications of CubeSat, which was developed by Stanford University, the USA, and California Polytechnic State University, the USA, back in 1999. CubeSat specifications basically call for a 10cm cube that fills up to a volume of exactly one litre, and can weigh up to 1.33kg.

Equipped with cameras, sensors and radios, the project is built to be an open source platform that will enable others to run experiments in space without having to send up a satellite of their own.

ardusat.com is a very good source for learning resources. Set up with the aim to expand science, technology, engineering and maths (STEM) innovation to space, it